

Esteban Armas Vega

Cybersecurity Researcher | Threat Intelligence | Applied AI | EU R&D Projects

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Professional Profile

Cybersecurity researcher with 10+ years of experience across applied research, software development, and European R&D projects. Specialized in cyber threat intelligence, threat analysis, secure information sharing, OSINT-driven enrichment, and AI-assisted cybersecurity solutions. Proven track record delivering technical work packages, research publications, prototypes, and project proposals in multi-partner international environments. Strong blend of academic depth and hands-on engineering across Python, MISP, machine learning, cloud-native tooling, and security analytics.

Core Strengths

- Cyber Threat Intelligence (CTI)
- Threat Hunting & Threat Analysis
- MISP-based intelligence sharing
- OSINT collection and enrichment
- Applied AI for cybersecurity
- Security data pipelines and automation
- Research writing and EU deliverables
- Horizon Europe / H2020 collaboration
- Malware and phishing analysis support
- Security analytics and anomaly detection

Professional Experience

Researcher

ATOS Spain / Eviden, Research & Innovation

Nov 2020 – Present

Madrid, Spain

- Design and develop cybersecurity solutions for European R&D projects, with emphasis on CTI enrichment, intelligence sharing, and security automation.
- Work on secure information exchange mechanisms for sensitive cyber data, using MISP as a core platform for structured intelligence collaboration.
- Research and implement frameworks, APIs, and analytics pipelines to enrich threat intelligence with OSINT and contextual data.
- Contribute to technical deliverables, architecture definition, partner coordination, and proposal writing for Horizon Europe initiatives.

Researcher

Complutense University of Madrid, GASS Research Group

Nov 2016 – Feb 2022

Madrid, Spain

- Conducted R&D in EU-funded cybersecurity projects spanning multimedia forensics, SDN security, malware and phishing detection, and applied machine learning.
- Developed computer vision algorithms for image and video forgery detection and contributed to a multimedia forensic analysis platform.
- Designed AI-based techniques to identify attack patterns in software-defined networks and supported scientific publications in high-impact venues.
- Participated in the preparation of European project proposals and collaborative technical reporting.

Developer

ESRI Geosystems

Nov 2015 – Oct 2016

Spain

- Built web and mobile applications combining ESRI technologies with open-source tools.
- Developed custom solutions using Python, JavaScript, Swift, Objective-C, C#, Spring, and MongoDB.
- Explored integration opportunities between geospatial platforms and emerging open-source technologies.

IT Manager

Armas Vega Asociados S.A.

Sep 2009 – Sep 2014

Quito, Ecuador

- Led software development and network deployment for business operations, including inventory, accounting, and enterprise management systems.
- Designed and managed office network infrastructure, servers, and web-based project management applications.

Selected Research & Innovation Projects

- **CYDERCO** – Threat intelligence sharing, enrichment, contextualization, and sighting support (2026 - On Going).
- **SAFE** – Security AI for Enhanced SOC (2027 - On Going).
- **CYLCOMED** – Cybersecurity toolbox for connected medical devices under Horizon Europe (2025 - Completed).
- **ELECTRON** – Resilience and self-healing for electrical power nanogrids (2025 - Completed).
- **HEROES** – Technologies to combat child sexual exploitation and human trafficking (2024 - Completed).
- **GEIGER** – Cybersecurity support and awareness tools for SMEs (2024 - Completed).
- **CyberSANE, CyberSec4Europe, CONCORDIA, RAMSES, SELFNET, CRYPTACUS.**

Education

PhD in Computer Science Complutense University of Madrid Thesis: <i>Passive Techniques for Detecting Manipulations in Digital Images</i>	Nov 2020
M.Sc. in Computer Science Complutense University of Madrid Thesis: <i>Tools for Dynamic Analysis of Vulnerabilities in Web Applications</i>	Oct 2016
B.Sc. in Informatics Engineering Technological University of Havana “José Antonio Echeverría”	Jul 2009
Scholarship Two-year Master Scholarship from the Secretariat of Higher Education of Ecuador	Sep 2014

Selected Publications

- **A Shared Cyber Threat Intelligence Solution for SMEs**, *Electronics*, 2021.
- **Copy-Move Forgery Detection Technique Based on Discrete Cosine Transform Blocks Features**, *Neural Computing and Applications*, 2020.
- **Image Tampering Detection by Estimating Interpolation Patterns**, *Future Generation Computer Systems*, 2020.
- **Passive Image Forgery Detection Based on the Demosaicing Algorithm and JPEG Compression**, *IEEE Access*, 2020.
- **Digital Images Authentication Technique Based on DWT, DCT and Local Binary Patterns**, *Sensors*, 2018.

Technical Skills

Programming	Python, Java, C#, JavaScript, Ruby, Swift, PHP, C, C++, SQL, Maude, Prolog
Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn, Caffe
DevOps / Platforms	Git, Docker, Kubernetes, Prometheus, Grafana, Linux, Windows, macOS, AWS, Azure, GCP
Security & CTI	MISP, YARA, Cuckoo Sandbox, OSINT tools, CTI workflows, secure data sharing

Languages

Spanish (native) | English (C1 equivalent)